

July 02nd 2024

Graphene synthesis project update

1. Biochar synthesis

- Overall seven syntheses finished with hemp fibers from BAFA neu GmbH
 - three syntheses done with 0.02M sulfuric acid
 - first product not representative, not used further
 - second and third batch converted (see below)
 - four syntheses done with 0.10M sulfuric acid
 - very similar observations and outputs
 - products of these have been combined/homogenized
 - part of the combined lots has been converted
- Canadian Rockies Hemp arrived and cleared customs
 - thirteen synthesis batches finished (21.6 g product in total), seven of these (11.2 g) were combined and homogenized
 - experiment with higher sulfuric acid concentration in progress
- (Re)-Activation of larger vessel completed, first production batch in preparation

2. Biochar analysis

- Sample of combined biochar (milled, MB2946X_gem) analyzed by SEM and Raman for reference

3. Graphene synthesis

- two experiments finished with steep heating ramp (20-30 °C/min)
 - biochar ground manually for first experiment (TB1131)
 - laboratory mill used for second experiment (TB1132)
 - products not yet characterized
 - no serious handling issues
- two experiments performed with much gentler heating ramp (3 °C/min, TB1133/1134)
 - better handling but no other apparent difference
- one experiment performed at 5 °C/min (MB2946)
- 2 g of combined biochar milled with 2 g of KOH and used for several batches on 0.25 g scale
 - 800 °C, 1 h: minimal yield (MB2947A)
 - 800 °C, 2 h: minimal yield (MB2947B)

- 800 °C, 3 h: no output (MB2947C)
 - Inertion apparently insufficient and material likely incinerated
- 800 °C, 3 h repeat, much higher Ar stream: 38 mg output obtained (MB2947D)
- 800 °C, 1 h, nitrogen inertion: 60 mg output (TB1135-1)
- 800 °C, 1 h, increased nitrogen inertion: 70 mg output (TB1135-2)

- 2 g of combined biochar milled with 2 g of KOH and used for several batches on 0.25 g scale
 - 800 °C, 2 h, inertion equivalent to TB1135-2: 70 mg output (MB2952B)
 - 800 °C, 3 h, inertion equivalent to TB1135-2: 50 mg output (MB2952C)
 - smoldering observed upon contact of the cold material with air (see video)
 - 800 °C, 3 h, inertion equivalent to TB1135-2: 60 mg output (MB2952C2)
 - material wetted with water before/during removal from the apparatus

- Oxygen sensor shows complete inertion of apparatus within 3 minutes at selected nitrogen stream (equivalent to TB1135-2)

- 2 g of combined biochar milled with 3 g of KOH and used for several batches on 0.25 g scale
 - 800 °C, 1 h, standard inertion: Quartz tube shattered, only 7 mg output (MB2955A)
 - replacement tubes received, additional experiments in preparation

4. Graphene analysis

- SEM (and Raman) results available for
 - TB1133, MB2946A, MB2947D, TB1135-1, TB1135-2, MB2952B, MB2952C, MB2952C2, MB2955A (step 2 products)
 - 900561-500MG (Sigma-Aldrich reference)
 - 924458-500MG (Sigma-Aldrich reference “Single-layer graphene sheets for battery, Bio-sourced”)

- Discussion with potential service provider (TEM, BET) close to completion

5. Currently open questions

Experiment overview

Experiment	Reactor	Raw material	Acid	T	t	pressure	Wash	Output	Experiment	Qty	Base	grinding	Gas	T (rate)	T (max)	t	Wash	Drying	Output				
MB2928	AV1	6.0 g BAFA neu Hemp Fibre VF	100.0 g 0.19% 0.02 M Sulfuric acid	180 °C	22.5 h	8.7 bar -> 10.4 bar	742 g Water	2.6 g	->														
Comment	Fibres not completely covered by liquid																						
MB2933	AV1	6.1 g BAFA neu Hemp Fibre VF	174.0 g 0.19% 0.02 M Sulfuric acid	180 °C	24 h	8.7 bar -> 9.5 bar	647 g Water	1.2 g	->	TB1131	1.2 g	1.2 g KOH (90%)	manual	Ar	20-27 °C/min	790 °C	1 h	20 g 10% HCl	100 °C N2 stream	0.16 g			
Comment	dark crusts on glassware, metal and liquid surface removed before filtration									Comment	ground biochar: brown powder, strong electrostatic charge												
MB2935	AV1	6.0 g BAFA neu Hemp Fibre VF	172.9 g 0.19% 0.02 M Sulfuric acid	180 °C	23 h	8.8 bar -> 9.5 bar	500 g Water	1.3 g	->	TB1132	1.3 g	1.3 g KOH (90%)	mill (2 min)	Ar	20-29 °C/min	792 °C	1 h	20 g 10% HCl	100 °C N2 stream	0.3 g			
Comment	dark crusts on glassware, metal and liquid surface removed before filtration									Comment	ground biochar: brown powder									(+ water)	(atm. Pressure)		
MB2936	AV1	6.0 g BAFA neu Hemp Fibre VF	172.4 g 0.96% 0.10 M Sulfuric acid	180 °C	23 h	8.8 bar -> 10.6 bar	589 g Water	0.8 g	->	TB1133	1.0 g	1.0 g KOH (90%)	mill (1 min)	Ar	3 °C/min	785 °C	1 h	20 g 10% HCl	100 °C N2 stream	0.3 g			
Comment	dark crusts on glassware, metal and liquid surface removed before filtration									Comment	ground biochar: brown powder; see temperature trend									(+ water)	(atm. Pressure)		
MRa225	AV1	6.0 g BAFA neu Hemp Fibre VF	153.8 g 0.96% 0.10 M Sulfuric acid	180 °C	24 h	? -> 10.5 bar	500 g Water	0.9 g		TB1134	1.0 g	1.0 g KOH (90%)	mill (2 min)	Ar	3 °C/min	771 °C	1 h	20 g 10% HCl	100 °C N2 stream	0.3 g			
Comment	dark crusts on glassware, metal and liquid surface removed before filtration									Comment	ground biochar: brown powder; see temperature trend									(+ water)	(atm. Pressure)		
MRa228	AV1	6.3 g BAFA neu Hemp Fibre VF	154.0 g 0.96% 0.10 M Sulfuric acid	180 °C	22.5 h	8.7 bar -> 11.0 bar	500 g Water	0.9 g															
Comment	dark crusts on glassware, metal and liquid surface removed before filtration																						
MRa231	AV1	6.1 g BAFA neu Hemp Fibre VF	168.8 g 0.96% 0.10 M Sulfuric acid	180 °C	23.5 h	8.0 bar -> 10.5 bar	500 g Water	0.8 g															
Comment	dark crusts on glassware, metal and liquid surface removed before filtration																						

Experiment	Reactor	Raw material	Acid	T	t	pressure	Wash	Output	Experiment	Qty	Base	grinding	Gas	T (rate)	T (max)	t	Wash	Drying	Output				
MB2937	AV1	9.4 g Canadian Rockies Hemp	179.3 g 0.96% 0.10 M Sulfuric acid	180 °C	24 h	8.6 bar -> 10.9 bar	569 g Water	1.6 g	MB2946	1.0 g	1.0 g KOH (90%)	mill (1 min)	Ar	5 °C/min	770 °C	1 h	22 g 10% HCl	100 °C N2 stream	0.3 g				
Comment	very low amount of foreign material observed and removed during filtration									Comment	ground biochar: brown powder; see temperature trend									(+ water)	(atm. Pressure)		
MB2938	AV1	8.9 g Canadian Rockies Hemp	175.0 g 0.96% 0.10 M Sulfuric acid	180 °C	23.5 h	9.2 bar -> 13.2 bar	577 g Water	1.4 g	MB2947A	0.25 g	0.25 g KOH (90%)	mill (1 min)	Ar	3 °C/min	803 °C	1 h	10 g 10% HCl	40 °C N2 stream	<0.01 g				
Comment	very low amount of foreign material observed and removed during filtration									Comment	ground biochar: brown powder; see temperature trend									(+ water)	210 mbar		
MB2940	AV1	9.2 g Canadian Rockies Hemp	177.1 g 0.96% 0.10 M Sulfuric acid	180 °C	22.5 h	8.7 bar -> 10.6 bar	555 g Water	1.5 g	MB2947B	0.25 g	0.25 g KOH (90%)	mill (1 min)	Ar	3 °C/min	804 °C	2 h	10 g 10% HCl	40 °C N2 stream	<0.01 g				
Comment	very low amount of foreign material observed and removed during filtration									Comment	ground biochar: brown powder; see temperature trend									(+ water)	210 mbar		
MB2942	AV1	8.9 g Canadian Rockies Hemp	177.6 g 0.96% 0.10 M Sulfuric acid	180 °C	23 h	8.3 bar -> 13.5 bar	578 g Water	1.6 g	MB2947C	0.25 g	0.25 g KOH (90%)	mill (1 min)	Ar	3 °C/min	806 °C	3 h							
Comment	low amount of foreign material observed and removed during filtration									Comment	ground biochar: brown powder; see temperature trend									Batch discarded			
MB2943	AV1	9.8 g Canadian Rockies Hemp	178.2 g 0.96% 0.10 M Sulfuric acid	180 °C	24 h	8.3 bar -> 12.7 bar	611 g Water	1.7 g	MB2947D	0.25 g	0.25 g KOH (90%)	mill (1 min)	Ar	3 °C/min	810 °C	3 h	10 g 10% HCl	100 °C N2 stream	38 mg				
Comment	low amount of foreign material observed and removed during filtration									Comment	ground biochar: brown powder; see temperature trend									(+ water)	(atm. Pressure)		
MB2944	AV1	9.0 g Canadian Rockies Hemp	178.4 g 0.96% 0.10 M Sulfuric acid	180 °C	24 h	8.4 bar -> 9.7 bar	593 g Water	1.8 g	TB1135-1	0.25 g	0.25 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	1 h	14 g 10% HCl	100 °C N2 stream	0.06 g				
Comment	low amount of foreign material observed and removed during filtration									->	Comment	ground biochar: brown powder; see temperature trend									(+ water)	(atm. Pressure)	
MB2945	AV1	9.1 g Canadian Rockies Hemp	177.7 g 0.96% 0.10 M Sulfuric acid	180 °C	24 h	8.7 bar -> 9.8 bar	6 g Water	1.6 g	TB1135-2	0.25 g	0.25 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	1 h	14 g 10% HCl	100 °C N2 stream	0.07 g				
Comment	low amount of foreign material observed and removed during filtration									Comment	ground biochar: brown powder; see temperature trend									(+ water)	(atm. Pressure)		

Experiment	Reactor	Raw material	Acid	T	t	pressure	Wash	Output	Experiment	Qty	Base	grinding	Gas	T (rate)	T (max)	t	Wash	Drying	Output
MB2948	AV1	9.4 g Canadian Rockies Hemp	175.1 g 0.96% 0.10 M Sulfuric acid	180 °C	25 h	9.0 bar -> 12.6 bar	475 g Water	1.7 g											
Comment	low amount of foreign material observed and removed during filtration																		
MB2949	AV1	9.2 g Canadian Rockies Hemp	174.3 g 0.96% 0.10 M Sulfuric acid	180 °C	23.5 h	8.4 bar -> 10.2 bar	538 g Water	1.8 g											
Comment	low amount of foreign material observed and removed during filtration																		
MB2950	AV1	9.1 g Canadian Rockies Hemp	170.5 g 0.96% 0.10 M Sulfuric acid	180 °C	23 h	-> 11.1 bar	534 g Water	1.6 g											
Comment	low amount of foreign material observed and removed during filtration																		
MB2951	AV1	9.3 g Canadian Rockies Hemp	178.5 g 0.96% 0.10 M Sulfuric acid	180 °C	24 h	9.1 bar -> 11.2 bar	575 g Water	1.8 g											
Comment	low amount of foreign material observed and removed during filtration																		
MRa255	AV1	9.2 g Canadian Rockies Hemp	170.8 g 0.96% 0.10 M Sulfuric acid	180 °C	23.5 h	8.1 bar -> 12.8 bar	488 g Water	1.7 g											
Comment	some foreign material observed and removed during filtration																		
MB2953	AV1	9.7 g Canadian Rockies Hemp	177.1 g 0.96% 0.10 M Sulfuric acid	180 °C	23.5 h	9.1 bar -> 10.4 bar	558 g Water	1.8 g											
Comment	low amount of foreign material observed and removed during filtration																		
MB2956	AV1	9.8 g Canadian Rockies Hemp	172.0 g 0.96% 0.10 M Sulfuric acid	180 °C	24 h	8.6 bar -> 12.5 bar	548 g Water												
Comment	low amount of foreign material observed and removed during filtration																		
MB2957	AV1	9.1 g Canadian Rockies Hemp	176.8 g 1.50% 0.15 M Sulfuric acid	180 °C		->	Water												
Comment																			